

GradCafe Admissions Analysis

Module 3 - SQL Query Results

The SQL shown below matches the uploaded final query_data.py. Results reflect the final console output after the tested data refresh.

Question 1

Question

How many entries in your database are from applicants who applied for Fall 2026?

Final Result

Fall 2026 applicant count: 11,874

SQL Query

```
SELECT COUNT(*)  
FROM applicants  
WHERE term = 'Fall 2026';
```

Explanation

The query filters the term field to Fall 2026 and counts every matching entry.

Question 2

Question

Among entries that provide a usable nationality classification, what percentage are international students?

Final Result

Percent international: 49.01%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE us_or_international = 'International'
    ) / NULLIF(
        COUNT(*) FILTER (
            WHERE us_or_international IN (
                'American',
                'International'
            )
        ),
        0
    )
FROM applicants;
```

Explanation

The numerator counts International entries. The denominator counts entries classified as either American or International, so missing or blank classifications are excluded.

Question 3

Question

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Final Result

Average GPA: 3.75

Average GRE Quantitative: 253.57

Average GRE Verbal: 159.94

Average GRE Analytical Writing: 7.52

SQL Query

```
SELECT
    AVG(gpa),
    AVG(gre),
    AVG(gre_v),
    AVG(gre_aw)
FROM applicants;
```

Explanation

PostgreSQL AVG ignores NULL values, so each column is averaged independently using the applicants who provide that particular metric.

Question 4

Question

What is the average GPA of American applicants who applied for Fall 2026?

Final Result

Average GPA of American Fall 2026 applicants: 3.76

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE term = 'Fall 2026'
      AND us_or_international = 'American'
      AND gpa IS NOT NULL;
```

Explanation

The query restricts entries to Fall 2026 American applicants with a non-missing GPA and calculates their average GPA.

Question 5

Question

What percentage of Fall 2025 entries are acceptances?

Final Result

Fall 2025 acceptance percentage: 54.43%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE status = 'Accepted'
    ) / NULLIF(COUNT(*), 0)
FROM applicants
WHERE term = 'Fall 2025';
```

Explanation

The denominator is all Fall 2025 entries, while the filtered numerator counts only entries with Accepted status.

Question 6

Question

What is the average GPA of accepted applicants who applied for Fall 2026?

Final Result

Average GPA of accepted Fall 2026 applicants: 3.75

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE term = 'Fall 2026'
      AND status = 'Accepted'
      AND gpa IS NOT NULL;
```

Explanation

The query keeps accepted Fall 2026 entries with a GPA and calculates the average of those GPA values.

Question 7

Question

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Final Result

JHU Computer Science master's applicant count: 6

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE degree = 'Masters'
      AND LOWER(program) LIKE '%computer science%'
      AND (
          LOWER(university) LIKE '%johns hopkins university%'
          OR LOWER(university) = 'jhu'
      );
```

Explanation

Using the original downloaded fields, the query requires a Masters degree, a Computer Science program match, and either Johns Hopkins University or JHU in the university field.

Question 8

Question

Using the original downloaded fields, how many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown University, MIT, Stanford University, or Carnegie Mellon University?

Final Result

Original-field count: 1

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE term = 'Fall 2026'
      AND status = 'Accepted'
      AND degree = 'PhD'
      AND LOWER(program) LIKE '%computer science%'
      AND (
        LOWER(university) LIKE '%georgetown university%'
        OR LOWER(university) LIKE
          '%massachusetts institute of technology%'
        OR LOWER(university) = 'mit'
        OR LOWER(university) LIKE '%stanford university%'
        OR LOWER(university) LIKE
          '%carnegie mellon university%'
      );
```

Explanation

The query simultaneously filters the original fields by Fall 2026, Accepted status, PhD degree, Computer Science, and one of the four specified universities.

Question 9

Question

Repeat Question 8, but identify the university and program using `llm_generated_program` and `llm_generated_university` instead of the original downloaded university/program information.

Final Result

Original-field count: 1

LLM-field count: 1

Difference: +0

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE term = 'Fall 2026'
AND status = 'Accepted'
AND degree = 'PhD'
AND LOWER(llm_generated_program)
    LIKE '%computer science%'
AND (
    LOWER(llm_generated_university)
        LIKE '%georgetown university%'
    OR LOWER(llm_generated_university)
        LIKE '%massachusetts institute of technology%'
    OR LOWER(llm_generated_university) = 'mit'
    OR LOWER(llm_generated_university)
        LIKE '%stanford university%'
    OR LOWER(llm_generated_university)
        LIKE '%carnegie mellon university%'
);
```

Explanation

This repeats Question 8 but uses the LLM-generated program and university fields. Both approaches return 1 matching entry, so the difference is +0. For this subset, the LLM standardization did not change the final count.

Original Question 1

Question

What percentage of Fall 2026 entries are acceptances?

Final Result

Fall 2026 acceptance percentage: 40.40%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE status = 'Accepted'
    ) / NULLIF(COUNT(*), 0)
FROM applicants
WHERE term = 'Fall 2026';
```

Explanation

The denominator contains all Fall 2026 entries, and the numerator counts only Fall 2026 entries with Accepted status.

Original Question 2

Question

What is the average GPA of international applicants?

Final Result

Average GPA of international applicants: 3.76

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE us_or_international = 'International'
AND gpa IS NOT NULL;
```

Explanation

The query selects International applicants who provide a GPA and calculates the average of those GPA values.